

SUPERSTORE SALES ANALYTICS

Power BI Business Intelligence & Data Analytics Report

Project Detail	Information
Project	Superstore Sales Analytics
Business Domain	Retail / E-commerce
Tool	Microsoft Power BI
Dataset	Sample Superstore (Kaggle)
Analysis	Sales, Profit, Products, Customers, Regions and Discounts
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1. Executive Summary

This project presents an interactive Superstore Sales Analytics solution developed in Microsoft Power BI. The objective is to evaluate business performance across sales, profitability, products, customers, regions and discounting, and convert the analysis into practical business recommendations.

Based on the supplied dashboards, the business generated approximately \$2.30 million in sales and \$286.40 thousand in profit, with an overall profit margin of 12.47%. The dashboards show approximately 5K unique orders and 793 customers. Technology is the strongest category from a profitability perspective, while West is the strongest region by sales and profit.

A major profitability concern is Furniture: it contributes substantial sales but comparatively low profit. The discount-versus-profit analysis also indicates that higher discounting can weaken profitability.

2. Business Problem

Management needs a clear, interactive view of where revenue and profit are generated, which products and customers are most valuable, which regions perform best, and whether discounting is affecting profitability.

- Measure overall sales, profit, orders, customers and profitability.
- Identify high-performing and low-performing products and categories.
- Understand customer-segment and customer-level performance.
- Compare regional and geographic performance.
- Investigate the relationship between sales, profit and discount.
- Provide actionable recommendations for improving profitability.

3. Dataset & Methodology

The project uses the Sample Superstore dataset and analyzes order, customer, product and geographic dimensions together with Sales, Quantity, Discount and Profit measures. The dashboard applies descriptive and diagnostic analytics through KPI cards, trends, bar charts, donut charts, scatter plots, tables and interactive filters.

Area	Examples
Orders	Order ID, Order Date, Ship Date, Ship Mode
Customers	Customer ID, Customer Name, Segment
Products	Product Name, Category, Sub-Category
Geography	Region, State, City
Measures	Sales, Quantity, Discount, Profit
KPIs	Unique Orders, Unique Customers, AOV, Profit Margin

4. Key Performance Indicators

KPI	Dashboard Value	Interpretation
Total Sales	\$2.30M	Overall revenue.
Total Profit	\$286.40K	Overall profit.
Unique Orders	~5K	Distinct orders shown in the dashboard.
Unique Customers	793	Distinct customers.
Total Quantity	~38K	Units sold.
Profit Margin	12.47%	Profit as a percentage of sales.
Average Order Value	458.61	Average sales per unique order.
Average Profit per Order	\$57.18	Average profit per unique order.

Validation note: use 'Unique Orders' when the measure is a DISTINCTCOUNT of Order ID. Similarly, 'Unique Customers' is clearer than 'Total Customer'.

5. Dashboard Analysis

5.1 Executive Dashboard

Provides the management overview through sales/profit KPIs, monthly trend, category, region, product, customer and shipping analysis.

Key interpretation: The page shows a broadly stable sales trend and identifies strong category and regional contributors.

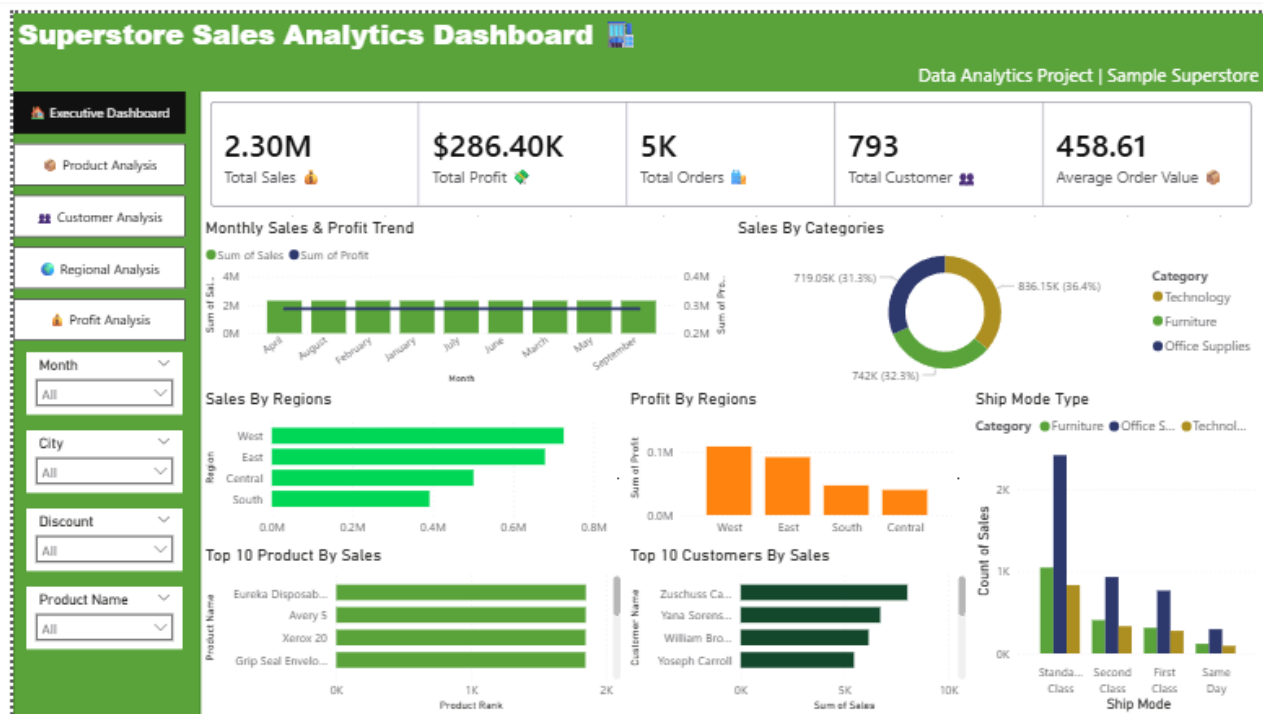


Figure: Executive Dashboard

5.2 Product Analysis

Compares category sales and profit, Top 10 products by sales and profit, Sales versus Profit and Discount Impact on Profit.

Key interpretation: Technology is the strongest category by profit. Comparing sales and profit rankings helps identify products that generate revenue efficiently.

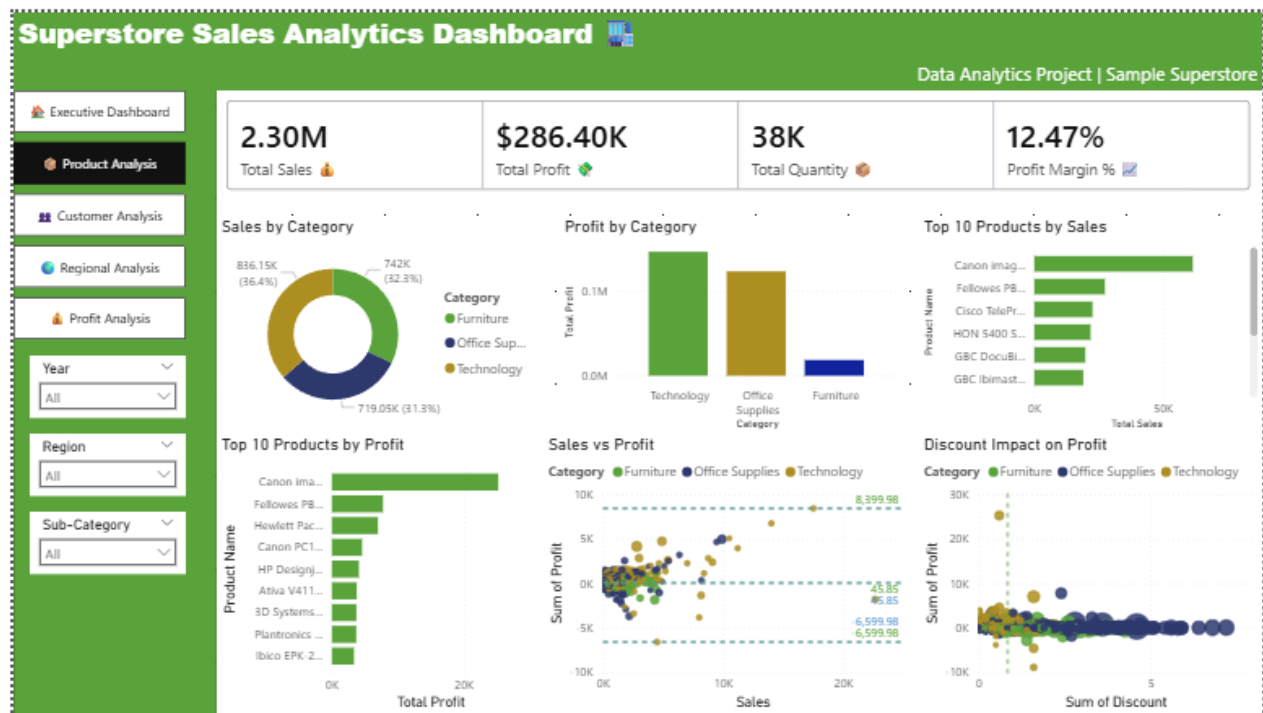


Figure: Product Analysis

5.3 Customer Analysis

Examines customer segments, top customers and customer-level sales versus profit.

Key interpretation: Consumer is the largest sales segment. Customer profitability should be evaluated alongside revenue to identify high-sales/low-profit accounts.

Superstore Sales Analytics Dashboard

Data Analytics Project | Sample Superstore



Figure: Customer Analysis

5.4 Regional Analysis

Compares the four regions, states, cities and segment-level performance.

Key interpretation: West is the strongest region by sales and profit. Central has weaker profitability relative to its sales contribution.

Superstore Sales Analytics Dashboard

Data Analytics Project | Sample Superstore

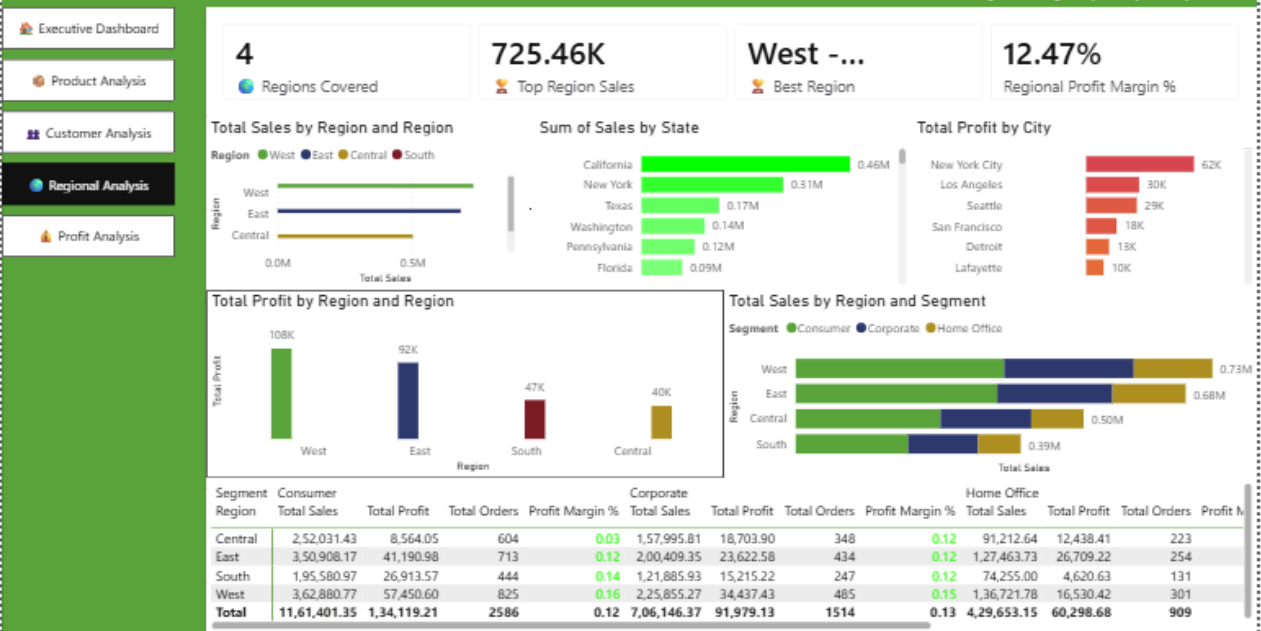


Figure: Regional Analysis

5.5 Profit Analysis

Focuses on profit, margin, loss-making products, category profitability, regional profit and discount/quantity relationships.

Key interpretation: Technology is the most profitable category, while discounting is an important profitability lever.

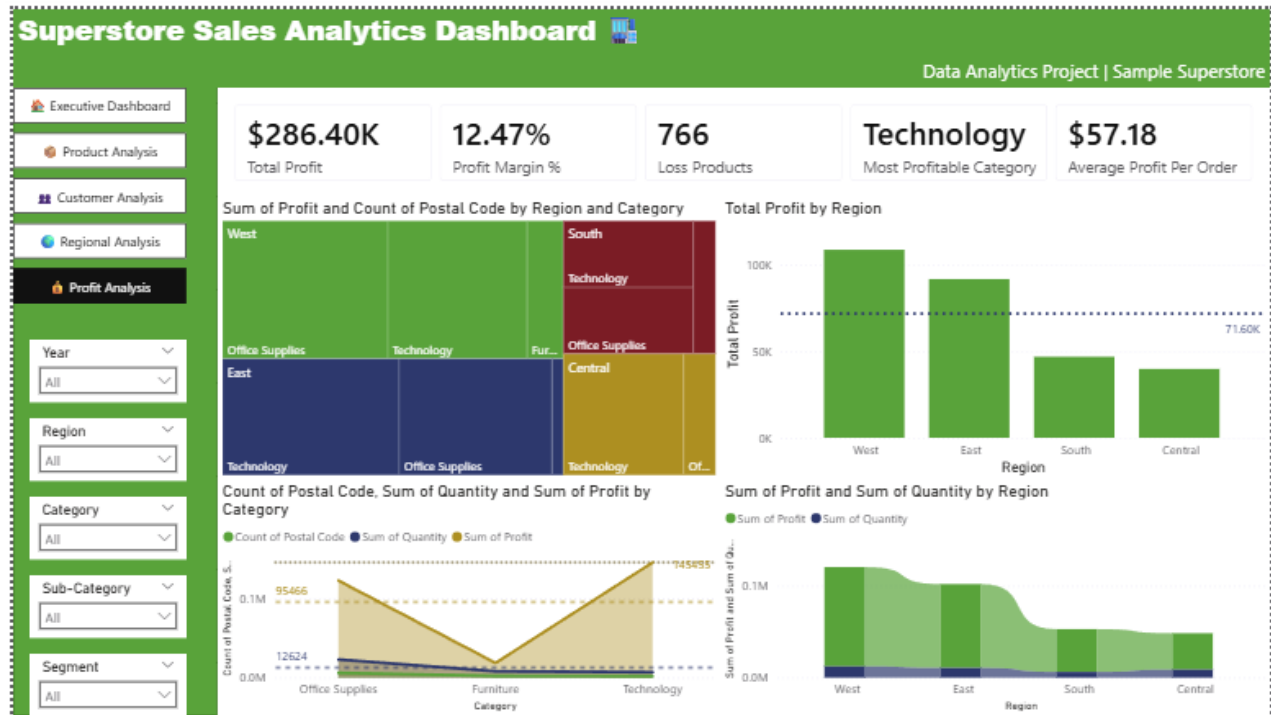


Figure: Profit Analysis

6. Category Analysis

Category	Approx. Sales	Approx. Share	Interpretation
Technology	\$836K	36.4%	Largest sales category and strongest profitability contributor.
Furniture	\$742K	32.3%	Strong sales but comparatively weak profit.
Office Supplies	\$719K	31.3%	Strong revenue contribution and solid profitability.

The key category insight is that revenue share and profitability are not identical. Furniture produces a substantial sales contribution but a much smaller profit contribution than Technology.

7. Regional Analysis

Region	Approx. Sales	Approx. Profit	Observation
West	\$725K	\$108K	Strongest overall performer.
East	\$679K	\$92K	Strong second performer.
Central	\$501K	\$40K	Lower profitability relative to sales.
South	\$392K	\$47K	Lower sales base with meaningful profit.

West is the strongest region in the dashboard. Central deserves attention because its sales contribution is relatively high while its profit contribution is comparatively low.

Important correction: the Regional Analysis page displays 12.47% as a regional profit-margin KPI. If 12.47% is the overall business margin, rename the card to 'Overall Profit Margin' or calculate the selected region's actual margin before publishing.

8. Customer Analysis

- Consumer is the largest customer segment by sales.
- Customer sales should be evaluated together with profit and margin.
- Top-customer analysis can support retention and account-management decisions.
- High-sales but low-profit customers should be investigated for discounts, product mix and order characteristics.

9. Discount & Profitability Analysis

The Discount Impact on Profit visual indicates that higher discount levels are associated with weaker profit outcomes and more loss-making observations. The appropriate business response is not to eliminate discounts, but to optimize discounts by product margin, category, customer segment and business objective.

10. Key Business Findings

- Approximately \$2.30M in sales and \$286.40K in profit are generated, with an overall profit margin of 12.47%.
- Technology is the strongest category from a profitability perspective.
- Furniture has relatively strong sales but weak profit and should be investigated.
- West is the strongest region by sales and profit.
- Central has a profitability gap relative to its sales contribution.
- Consumer is the largest customer segment.
- High sales do not always translate into proportionally high profit.
- Higher discounting is associated with weaker profitability.

11. Business Recommendations

Review Furniture profitability

Analyze product-level margins, discounting, shipping and product mix.

Optimize discounts

Introduce margin-based discount thresholds rather than relying only on revenue growth.

Learn from West

Compare product mix, customer mix and discount behavior with weaker regions.

Manage customer profitability

Use sales, profit, margin and order frequency together.

Monitor loss-making products

Review products with negative aggregated profit regularly.

Use profit with sales

Evaluate growth using both revenue and profitability to avoid low-quality sales growth.

12. Technical Skills Demonstrated

- Power BI dashboard development
- DAX KPI and measure design
- Power Query / data preparation
- Sales and profitability analysis
- Customer and product segmentation
- Regional analysis
- Interactive slicers and filtering
- Trend and scatter-plot analysis
- Business storytelling and recommendations

13. Limitations & Validation

- The report is based on the supplied dashboard screenshots; the underlying Power BI model and DAX measures were not independently audited.
- Some product and customer names are truncated in the screenshots.
- The definition behind the 'Loss Products' KPI should be verified before publication.
- The 12.47% Regional Profit Margin label should be corrected if it represents the overall margin.
- The analysis describes observed relationships and does not establish causality.

14. Conclusion

The Superstore Sales Analytics project is a strong end-to-end Power BI business intelligence example. It combines executive KPIs with product, customer, regional and profitability analysis and provides actionable findings rather than simply displaying charts.

The strongest business message is that revenue alone does not determine performance. Technology and West are major contributors, while Furniture and Central present profitability opportunities. Discount management is another important lever for improving margins.

15. Portfolio Project Summary

Portfolio Element	Recommended Text
Project Title	Superstore Sales & Profitability Analytics
Tools	Power BI, DAX, Power Query
Domain	Retail / E-commerce
Objective	Identify sales and profitability drivers across products, customers and regions.
Key Finding	Technology and West lead performance; Furniture and Central require profitability attention.
Business Value	Supports pricing, discount, product, customer and regional decisions.